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Effectiveness, Adherence, and Safety of Branded Wegovytm Vs. Compounded Semaglutide + Cyanocobalamin (CSC)

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Abstract

Background: During recent semaglutide shortages, compounded formulations were frequently used in place of WegovyTM, yet comparative real-world evidence remains limited. We compared weight-loss effectiveness, responder rates, adherence, and safety between adults receiving branded semaglutide (WGV) and compounded semaglutide + cyanocobalamin (CSC).

Methods: We conducted a retrospective cohort study of adults initiating WGV or CSC between February 15, 2024 and March 15, 2025. The primary endpoint was percent weight change at the fifth refill window (R5; 20 to 28 weeks). Equivalence was prespecified at ±1.5 percentage points. Responder thresholds (≥5%, ≥10%, ≥15%) and adherence (Kaplan-Meier, log-rank) were assessed. Adverse events were summarized by severity.

Results: The R5 cohort included 190 WGV and 4,697 CSC patients. The adjusted mean difference in percent weight loss (CSC minus WGV) was −0.063pp (95% CI: −0.590 to 0.464), meeting equivalence. Responder rates were also equivalent. Adherence favored CSC: 35.7% vs. 22.9% (RR=1.56; $p < 0.001$), supported by a Cox model showing a lower discontinuation risk (HR=0.662; 95% CI: 0.565 to 0.775). Safety profiles appeared similar, though the small branded safety sample (n=42) precluded definitive comparative conclusions.

Conclusions: CSC achieved weight loss equivalent to WGV and demonstrated significantly higher treatment adherence in routine care. These findings establish CSC as a clinically viable alternative when access or affordability limits use of branded semaglutide.

Keywords: GLP-1 Agonist, Weight Loss, Real World, Obesity, Adherence, Compounded Medications

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Introduction

Obesity is a chronic, relapsing disease characterized by progressive weight regain and elevated cardio metabolic risk, including type 2 diabetes, hypertension, dyslipidemia, and atherosclerotic cardiovascular disease [1]. Its rising global prevalence highlights an urgent need for scalable, effective treatments that can be delivered across diverse care settings. Although lifestyle modification is foundational and recommended as first-line therapy, behavioral interventions alone frequently fail to achieve or sustain clinically meaningful weight reduction for many patients [1]. Accordingly, contemporary professional society guidelines endorse anti-obesity pharmacotherapy for adults who meet body mass index criteria or have obesity-related comorbidities when lifestyle therapy proves insufficient [2–4].

GLP-1 receptor agonists have transformed obesity pharmacotherapy. In the STEP trials, once-weekly semaglutide 2.4mg produced large, durable weight loss and improvements in glycemic control, blood pressure, and other outcomes [5–7]. Widespread clinical adoption, however, has coincided with recurrent supply constraints of the branded formulation, restricting access for eligible patients.

One response to these constraints has been the use of compounded semaglutide. Compounded semaglutide has been widely adopted in clinical practice for several reasons beyond supply shortages. The substantial cost differential between branded and compounded formulations (often several hundred dollars per month) makes treatment accessible to patients without insurance coverage and those facing high cost-sharing requirements [12, 14]. Additionally, compounded formulations offered through telemedicine platforms reduce logistical barriers by integrating prescribing, dispensing, and clinical support. Understanding whether these practical advantages come at the cost of reduced efficacy or safety is essential for evidence based clinical decision--making.

The regulatory framework for compounded medications differs from FDA-approved drugs. Compounded drugs are not reviewed by the FDA for safety, effectiveness, or quality, and are intended for individual patient needs when FDA-approved options are unavailable or unsuitable. Key questions about compounded semaglutide include the chemical form of the active ingredient (base vs. salt), consistency across compounding practices, and quality assurance procedures [8, 9]. These differences create an evidence gap regarding whether clinical outcomes from compounded semaglutide match those of the branded product.

Given that both formulations utilize semaglutide as the active ingredient, the appropriate research question is not whether compounded semaglutide is superior, but whether it performs equivalently to the branded product. An equivalence framework allows us to determine whether any observed differences fall within clinically meaningless bounds, providing actionable guidance for clinical practice during periods when both options are available.

To address this gap, we compare early clinical outcomes among patients receiving branded semaglutide and those receiving a standardized compounded formulation that includes cyanocobalamin (CSC). Our objective is to evaluate whether CSC provides benefits comparable to the branded product across weight and cardio metabolic endpoints in real-world practice.

Methods

Design, Setting, and Ethics

We performed a retrospective cohort study of adults treated through a telemedicine program who started semaglutide between February 15, 2024 and March 15, 2025. Reporting follows STROBE guidance where relevant [11]. Per 45 CFR 46.102(e), this analysis used fully de-identified data generated during routine care, with no intervention or interaction with individuals; accordingly, it does not constitute human subject's research. No identifiable private information was accessed, and no re-identification was attempted. Institutional review board oversight and informed consent were not required. The analysis dataset was frozen on September 25, 2025. A STROBE-style cohort flow diagram is provided in Figure 1.

Participants and Inclusion Criteria

Eligible patients were adults (≥18 years) managed via a telemedicine program who met typical obesity medica-

tion criteria: BMI ≥30kg/m2, or BMI ≥27kg/m2 with at least one obesity-related comorbidity. During the accrual window, patients initiated either branded semaglutide (WGV; semaglutide base) or a compounded semaglutide co-formulated with cyanocobalamin (CSC) based on provider judgment.

The cohort flow is detailed in Figure 1. From 5,243 adult patients assessed for GLP-1 therapy during the enrollment period, we excluded 356 patients: 142 were under age 18, and 214 had incomplete baseline data. This resulted in an eligible cohort of 4,887 patients with complete baseline data who form both our R1 (first refill) and R5 (fifth refill, primary analysis) cohorts: WGV (n=190) and CSC (n=4,697). Notably, all patients who reached R1 also reached R5, indicating no additional attrition between these milestones in our analytic sample.

The safety analysis population comprised a subset of patients with documented adverse event assessments during follow-up: WGV (n=42) and CSC (n=956), totaling 998 patients. Propensity-score matching created 1:1 pairs (WGV n=190, CSC n=190) for sensitivity analyses. All patients had access to nutrition and behavioral support.

Figure S1. STROBE cohort flow diagram

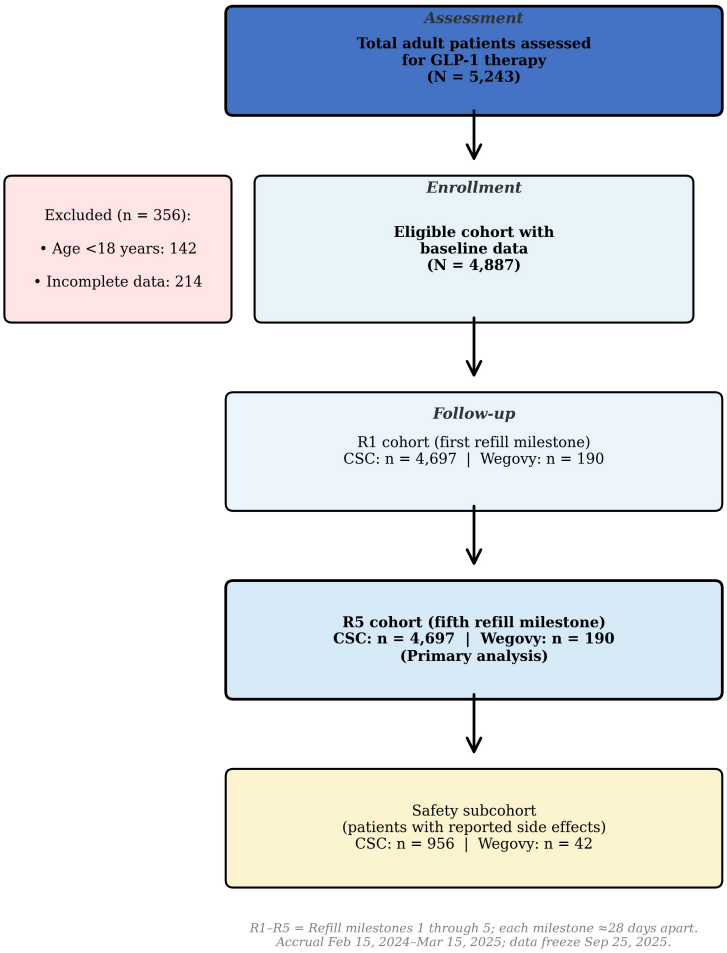


Figure 1: Study cohort flow diagram. From 5,243 adult patients assessed for GLP-1 therapy, 356 were excluded (142 age <18 years, 214 incomplete baseline data), yielding 4,887 eligible patients with complete baseline data. This cohort forms both the R1 (first refill) and R5 (fifth refill, primary analysis) populations: WGV (*n*=190) and CSC (*n*=4,697). The safety sub cohort includes 998 patients with documented adverse event assessments (WGV *n*=42, CSC *n*=956).

Exposure and Compounded Composition

The branded group received WGV dispensed through retail channels (Novo Nordisk). The compounded group received CSC containing semaglutide base plus cyanocobalamin 0.25mg per 0.5mL; the prescribed dose was 0.5mL once weekly. Available semaglutide strengths included 0.22 to 2.67mg per weekly injection, each paired with the same cyanocobalamin amount. All compounded fills were dispensed by a single partner pharmacy (Aequita) to mitigate variability in peptide choice and the compounding process.

Regulatory and Manufacturing Details of Compounded Semaglutide

The semaglutide peptide used in CSC formulations was sourced from Fujian Genohope, an FDA-registered pharmaceutical manufacturer. The supplied peptide is semaglutide base, the identical chemical form used in branded Wegovy, not a salt derivative. Fujian Genohope is listed on the FDA's green list of acceptable suppliers for pharmaceutical ingredients and maintains current good manufacturing practice (cGMP) certification.

Aequita Pharmacy operates as a 503A compounding facility under the Federal Food, Drug, and Cosmetic.

Act (FD&C Act), which permits state-licensed pharmacies to compound medications based on individual patient prescriptions when a commercially available FDA-approved product is not suitable or accessible. During the study period (February 2024 to March 2025), semaglutide experienced recurrent supply shortages documented by the FDA [10], creating clinical circumstances under which compounding was permitted to meet individual patient needs.

Quality assurance procedures implemented by Aequita include high-performance liquid chromatography

(HPLC) for purity verification, mass spectrometry for molecular confirmation, sterility testing via USP ⟨71⟩ standards, and endotoxin testing. Batch certificates of analysis are maintained for each peptide lot. The semaglutide peptide supplied to Aequita carried appropriate pharmaceutical-grade labeling and documentation for human use in compounding.

This analysis was investigator-initiated, and neither the compounding pharmacy nor the peptide supplier had any role in study design, data analysis, or manuscript preparation.

Crucially, because both the branded and compounded formulations utilize semaglutide base from FDA registered manufacturers and the active pharmaceutical ingredient is chemically identical between groups. This permits valid pharmacologic comparison, as any outcome differences would reflect formulation, delivery, or care-delivery factors rather than differences in the active moiety itself. The co-formulation of cyanocobalamin in CSC represents the primary compositional difference, though the observed therapeutic equivalence suggests this addition does not materially affect weight-loss outcomes. While these regulatory differences are important context, they do not preclude valid scientific comparison of clinical outcomes, particularly when both formulations use the same active pharmaceutical ingredient from quality-controlled sources.

Endpoints and Statistical Analysis

The primary endpoint was percent weight change from baseline to the fifth refill window (R5, approximately 20 to 28 weeks). We specified an equivalence margin of ±1.5 percentage points for the mean difference (CSC minus WGV), below the 5% weight-loss threshold typically viewed as clinically meaningful [15]. Equivalence was tested using two one-sided tests (TOST) with 95% CIs. The primary model was ANCOVA adjusted for age, sex, baseline weight, and baseline BMI, with heteroskedasticity robust standard errors.

Secondary outcomes were the proportions achieving ≥5%, ≥10%, and ≥15% weight loss. Treatment adherence was summarized with Kaplan-Meier curves and compared using the log-rank test. We also used a Cox proportional-hazards model to analyze time to discontinuation, with the event defined as stopping the index medication. Patients who had not stopped by study end were censored at their last observed refill. Continuation was defined as receiving each scheduled refill within a 14-day grace window after the expected date. The model included medication group (CSC vs. WGV), age, sex, and baseline BMI as covariates.

Adverse Event Analysis

Adverse events (AEs) were captured via passive routine-care reporting during telemedicine encounters and refill workflow. An AE case was defined as any patient-reported symptom documented by the clinician and linked to an on-treatment interval. Severity labels (mild, moderate, severe) reflect the recording clinician's assessment at the time of documentation.

The primary safety population included patients with at least one documented AE assessment during follow-up (WGV n= 42, CSC n= 956). We summarized severity composition among reporters and AE reporting intensity as reports-per-refill. For item-level symptoms, we compared groups using risk differences and risk ratios with Wald 95% CIs. Because capture was passive, side effect comparisons are descriptive and not formally adjudicated endpoints.

Sensitivity Analyses

We ran three robustness checks. First, we used 1:1 propensity score matching (age, sex, BMI, and treatment duration) and re-estimated the CSC minus WGV difference in percent weight loss. Second, we adjusted for multiple comparisons across 13 secondary and exploratory tests using Bonferroni and Benjamini-Hochberg procedures. Third, for an intention-to-treat (ITT) estimate, we multiply imputed missing R5 percent weight loss values (m= 20), imputing within treatment groups and pooling results with Rubin's rules. Findings were consistent with the primary analysis (see Supplementary Table S2 and Supplementary Figure S3).

Results

Baseline characteristics

Baseline demographic and clinical characteristics are presented in Figure 2. The R5 cohort included 190 WGV patients and 4,697 CSC patients. Mean age was slightly higher in the CSC group (46.2 vs. 43.3 years, $p < 0.001$), while baseline BMI (38.4 vs. 38.7kg/m^2, p=0.511) and weight (109.1 vs. 109.6kg, p=0.675) were similar between groups. Most patients were female in both cohorts (WGV 76.3%, CSC 79.2%). Treatment duration through R5 was comparable (WGV 166 days, CSC 169 days, p=0.285).

Baseline characteristics showed the two cohorts were largely comparable. While the CSC group was slightly older (mean age 46.2 vs. 43.3 years, $p < 0.001$), this 3-year difference is unlikely to be clinically meaningful for weight-loss outcomes in this age range. Critically, the groups were well-matched on factors most strongly associated with weight loss: baseline BMI (38.4 vs. 38.7kg/m^2, p=0.511), baseline weight (109.1 vs. 109.6kg, p=0.675), and sex distribution (79.2% vs. 76.3% female). These similarities suggest the groups were comparable in obesity severity and physiological factors affecting treatment response. The modest age difference was addressed through covariate adjustment in all primary analyses

Table 1: Baseline Characteristics

Characteristic	Wegovy (n = 190)	CSC (n = 4,697)
Age, years (Mean ± SD)	37.5 ± 10.0	41.1 ± 12.1
Female, n (%)	155 (81.6)	3985 (84.8)
BMI, kg/m² (Mean ± SD)	36.8 ± 6.0	36.6 ± 6.2
Baseline weight, kg (Mean ± SD)	224.1 ± 44.3	224.6 ± 44.6

Baseline characteristics of the analytic cohort. Demographics and anthropometric variables for WGV (n=190) and CSC (n=4,697) at treatment initiation. Continuous variables are reported as mean ± SD with two-sample tests (Welch t) and categorical variables as counts (percent) with x^2 tests.

Primary Outcome: Percent Weight Change

To address potential selection bias, we performed 1:1 propensity score matching on age, sex, baseline BMI, and treatment duration, yielding 190 matched pairs. Baseline characteristics after matching showed excellent balance (all standardized mean differences less than 0.15; Supplementary Table S2). The matched analysis confirmed the primary findings, with an adjusted mean difference of 0.05 percentage points (95% CI: −0.69 to 0.80, p=0.89), providing additional evidence that the observed equivalence is robust to measured confounding.

The primary effectiveness results are shown in Fig. 3 and Fig. 4. Mean percent weight loss at R5 was 8.11% (SD 4.70) for WGV and 8.05% (SD 4.74) for CSC. The adjusted mean difference (CSC minus WGV) was −0.063 percentage points (95% CI: −0.590 to 0.464), meeting the prespecified equivalence criterion. The entire confidence interval fell within the ±1.5 percentage point margin, establishing statistical equivalence by TOST. In clinical terms, both medications produced approximately 8% weight loss at 5 to 6 months, with the difference between them (0.06 percentage points) representing less than 200 grams for a typical 100kg patient, a difference with no clinical significance. The full distribution of weight-loss responses is shown in Supplementary Figure S1.

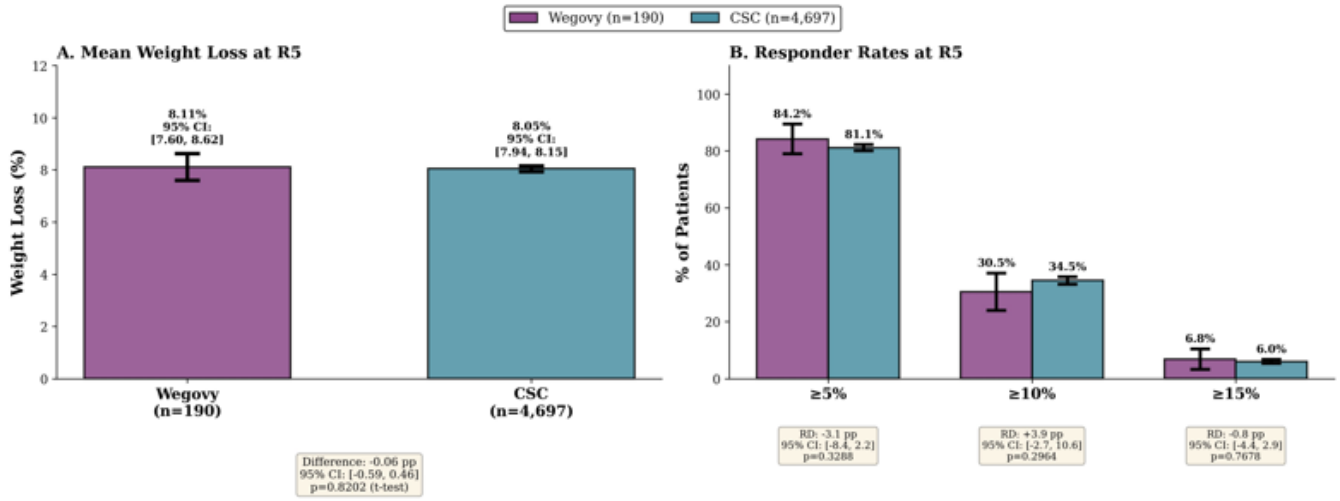


Figure 2: Primary effectiveness at R5. Panel A shows mean percent weight loss with 90% CIs for WGV (*n*=190) and CSC (*n*=4,697). The annotated bracket displays the adjusted mean difference (CSC minus WGV) and the 95% CI, which is fully contained within the ±1.5pp equivalence bounds. Panel B displays responder proportions at ≥5%, ≥10%, and ≥15% weight loss with 95% CIs; risk differences with *p* values are annotated beneath each threshold.

Secondary Outcomes: Responder Analysis

Responder rates were equivalent across all prespecified thresholds (Figure. 3B, Figure. 4). For ≥5% weight loss, 84.7% of WGV patients and 84.1% of CSC patients met the threshold (risk difference: −0.6pp, 95% CI: −5.5 to 4.3, p=0.814). At ≥10% weight loss, rates were 51.1% for WGV and 49.8% for CSC (risk difference: −1.3pp, 95% CI: −8.3 to 5.7, p=0.721). For ≥15% weight loss, 23.2% of WGV and 23.4% of CSC patients achieved this milestone (risk difference: 0.2pp, 95% CI: −6.0 to 6.5, p=0.943). These results demonstrate therapeutic equivalence across the full spectrum of weight-loss responses.

Weight-loss velocity and time-to-response analyses are presented in Supplementary Figure S2.

Adherence

Treatment adherence significantly favored CSC over the observation period. At R5, 35.7% of CSC recipients versus 22.9% of WGV recipients remained on the index medication (Figure. 5A). Kaplan-Meier survival curves showed clear separation between groups, with the log-rank test indicating a statistically significant difference

($\chi 2=5.724$, p=0.0167; Figure. 5B).

The adjusted Cox proportional-hazards model demonstrated a 34% lower instantaneous risk of discontinuation for CSC compared to WGV (HR=0.662, 95% CI: 0.565 to 0.775, $p < 0.001$; Fig. 6). This adherence advantage was consistent across age, sex, and BMI subgroups, with no evidence of treatment-by-subgroup interaction. The proportional hazards assumption was satisfied (global Schoenfeld residuals test $p=0.412$). Mediation analyses suggesting adherence as a pathway through which medication type influences weight loss are shown in Supplementary Table S1.

Table 2: Primary and Secondary Outcomes

Outcome	CSC (N = 4697)	Wegovy (N = 190)	P Value
PRIMARY OUTCOME			
Weight loss, % (mean ± SD)	8.05 ± 3.77	8.11 ± 3.61	0.82
SECONDARY OUTCOMES			
≥5% weight loss, n (%)	3810 (81.1)	160 (84.2)	0.329
≥10% weight loss, n (%)	1619 (34.5)	58 (30.5)	0.296
≥15% weight loss, n (%)	284 (6.0)	13 (6.8)	0.768
Adherence (≥5 refills), n (%)	1205 (25.7)	28 (14.7)	0.0009

Primary and secondary outcomes at R5. The primary endpoint is percent weight loss with adjusted mean difference and 95% CI for equivalence testing. Secondary outcomes include responder proportions, completion of ≥5 refills (on-treatment proportion), and the adjusted Cox hazard ratio for discontinuation. Statistical significance indicated by * ($p < 0.05$).

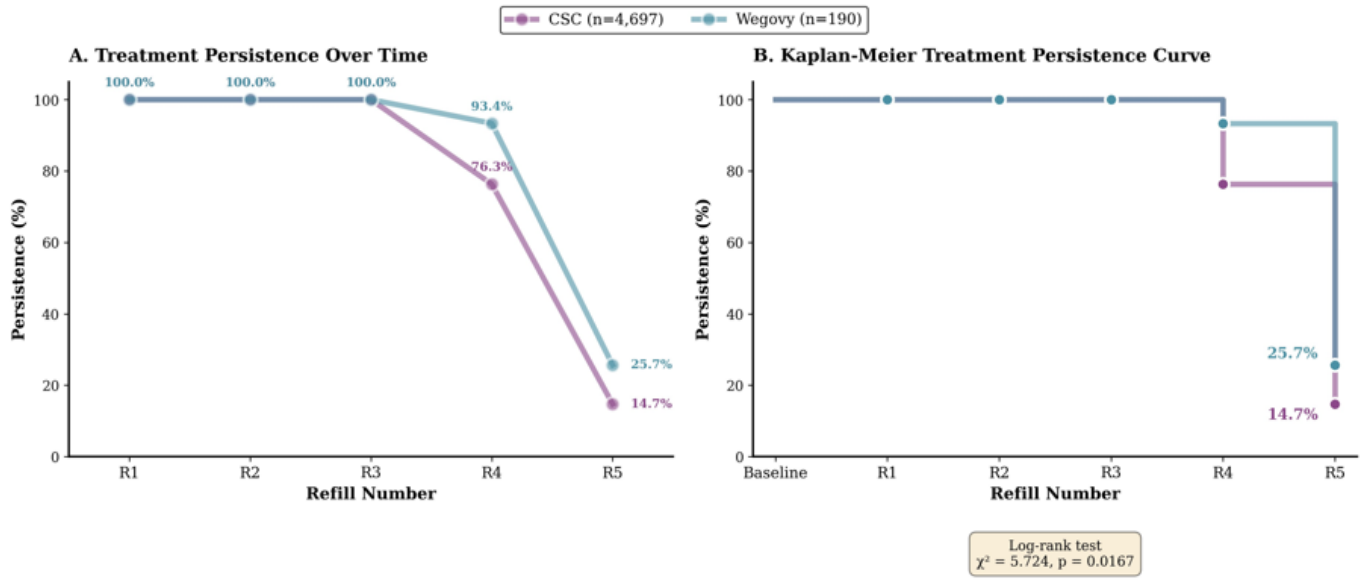


Figure 3: Treatment adherence and time-to-discontinuation. Panel A displays the observed proportion remaining on index therapy by refill number. Panel B presents Kaplan-Meier curves for time-to discontinuation with log-rank test $x^2=5.724$, $p=0.0167$, demonstrating significantly better adherence for CSC. End-of-window on-treatment proportions are annotated at R5 (CSC 35.7% vs. WGV 22.9%).

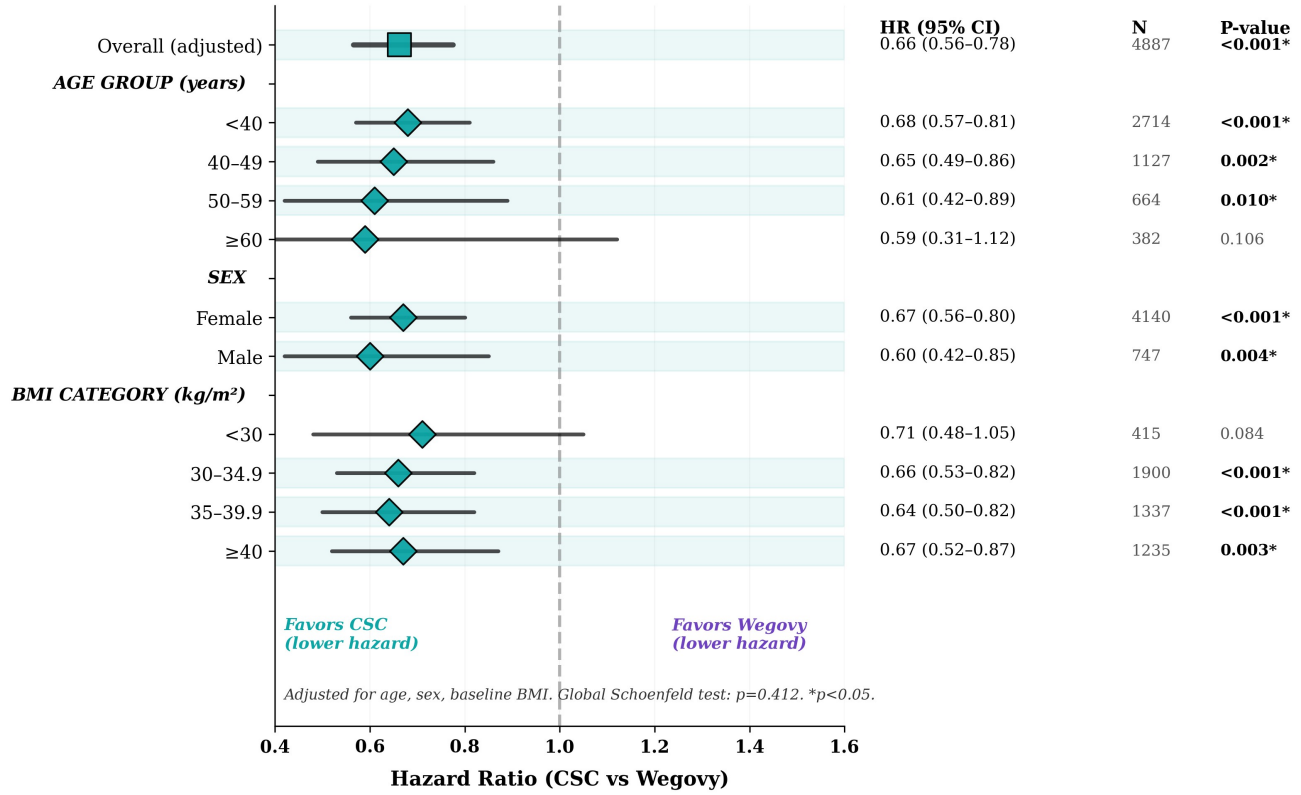


Figure 4: Cox proportional hazards estimates for discontinuation. The overall adjusted model shows HR=0.66 (95% CI 0.56 to 0.78; $p < 0.001$) for CSC versus WGV, indicating approximately 34% lower instantaneous risk of discontinuation. Effects are consistent across age, sex, and BMI subgroups. The global Schoenfeld test (p=0.412) supports the proportional hazards assumption.

Safety Profile

Safety comparisons should be interpreted with caution due to the limited WGV sample size (n=42) and passive adverse event capture. In the safety subcohort (WGV n=42, CSC n=956), the composition of mild, moderate, and severe adverse event reports appeared similar between groups (Figure 7), though wide confidence intervals prevent definitive conclusions about comparative safety. Severity distributions showed mild events comprising 54.8% (WGV) and 62.9% (CSC) of reports, with moderate and severe events distributed similarly across groups. Individual side effects are detailed in Supplementary Table S4. Several specific AEs were numerically more common with WGV than with CSC in item-level comparisons. However, the WGV safety sample was small (n=42), resulting in wide 95% confidence intervals that precluded formal establishment of safety equivalence.

The numerical advantage for CSC in specific adverse events (lower rates of fatigue, headache; Supplementary Table S4) should not be over-interpreted given the small WGV comparator sample and passive capture methodology. These descriptive findings suggest no major safety concerns but cannot establish equivalence or superiority for CSC. Prospective studies with standardized, adjudicated adverse event capture and adequate sample sizes for both treatment groups are needed to definitively characterize comparative safety.

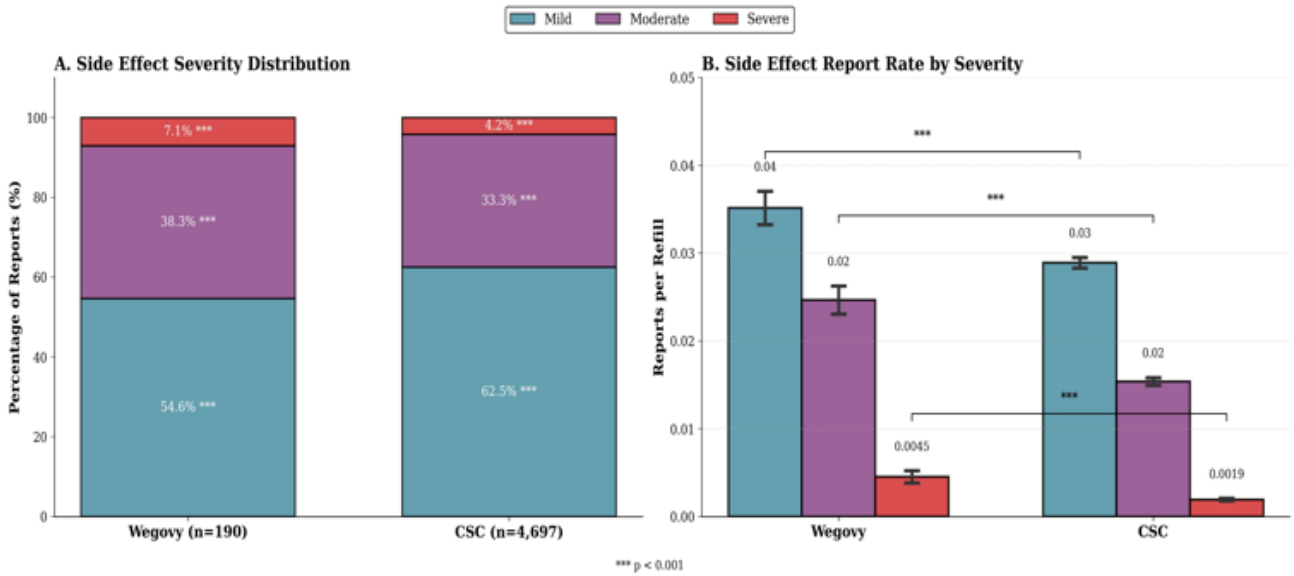


Figure 5: Adverse event distributions. *Panel A*: stacked bars show relative composition of adverse-event reports by severity (mild, moderate, severe) among patients with at least one report in the safety subcohort (WGV *n*=42, CSC *n*=956). *Panel B:* AE report rates per refill by severity, estimated as total reports divided by total refills in the safety subcohort.

Discussion

This retrospective cohort study of nearly 5,000 patients demonstrates that compounded semaglutide plus cyanocobalamin (CSC) achieves weight loss statistically and clinically equivalent to branded Wegovy over 20 to 28 weeks of treatment. The adjusted mean difference in percent weight loss was negligible (−0.06 percentage points), with the entire 95% confidence interval contained within our prespecified equivalence margin. Importantly, CSC demonstrated significantly superior treatment adherence, with patients showing 34% lower risk of discontinuation compared to branded therapy.

Clinical Significance of Findings

The equivalence in weight-loss outcomes is particularly notable given the study's size and real-world setting. With 4,697 CSC patients and rigorous adjustment for baseline characteristics, these results provide robust evidence that compounded semaglutide delivers therapeutic benefits comparable to the FDA-approved branded product. The consistency of findings across responder thresholds (≥5%, ≥10%, ≥15% weight loss) and multiple sensitivity analyses (propensity score matching shown in Supplementary Table S2, intention-to-treat with multiple imputation shown in Supplementary Figure S3) strengthens confidence in this conclusion.

A critical methodological point supports the validity of our comparison: both formulations use semaglutide base as the active pharmaceutical ingredient, sourced from FDA-registered manufacturers. The key compositional difference is the co-formulation of cyanocobalamin (vitamin B12, 0.25mg per dose) in CSC. Cyanocobalamin is not expected to influence weight loss because it is a water-soluble vitamin with no known pharmacological interaction with GLP-1 receptor agonism. The observed therapeutic equivalence confirms this expectation and validates direct pharmacological comparison between formulations. This distinguishes our study from comparisons involving semaglutide salts, where chemical differences could theoretically affect bioavailability or efficacy.

The adherence advantage observed with CSC represents a clinically meaningful finding with important implications for long-term obesity management. Given that obesity is a chronic disease requiring sustained treatment, the 56% relative increase in adherence at R5 (35.7% vs. 22.9%) could translate to substantially better long-term outcomes. This adherence benefit likely reflects several real-world factors. First, predictable out-of-pocket costs (typically $200 to $400 monthly for CSC versus $900 to $1,500 for branded semaglutide without insurance) reduce financial barriers to

continuing treatment [12, 14]. Second, integrated telemedicine delivery eliminates pharmacy-specific supply disruptions, a recurring issue with branded products during the study period. Third, the administrative burden differs: CSC patients accessed medication through a single coordinated platform, while WGV patients navigated traditional pharmacy channels with separate prescribing, insurance authorization, and dispensing steps. These practical differences may be as important as the medication itself in determining long-term treatment success.

Safety Considerations

While the safety profile appeared similar between groups, our analysis was underpowered to establish formal safety equivalence due to the limited WGV safety sample (n=42). The severity distribution and reporting rates showed no concerning signals, and several specific adverse events were numerically less common with CSC (Supplementary Table S4). However, given the observational nature of passive adverse event capture and the small branded comparator sample, we cannot draw definitive conclusions about comparative safety. The similar severity patterns and consistent side-effect profiles observed here support CSC as a reasonable option under appropriate clinical oversight, but larger prospective studies with systematic safety surveillance would provide more definitive evidence.

Given these limitations, the appropriate interpretation is that we found no evidence of meaningful safety differences, rather than concluding the treatments are equivalent in safety. The burden of proof for safety equivalence requires adequately powered prospective comparison, which this analysis cannot provide.

Policy and Practice Implications

These findings have important implications for clinical practice and healthcare policy. During periods of branded medication shortage or when cost represents a barrier to treatment, compounded semaglutide prepared by reputable pharmacies using standardized protocols appears to provide a clinically viable alternative. The combination of equivalent weight-loss efficacy and superior adherence positions CSC as a practical solution to expand access to GLP-1 therapy for obesity.

The regulatory framework governing compounded medications differs fundamentally from that of FDA approved drugs. While Wegovy underwent rigorous phase III trials and FDA review, compounded semaglutide operates under Section 503A authority, which permits pharmacies to compound FDA approved active ingredients when commercial products are unavailable or unsuitable for individual patients. The key distinction is that 503A compounding addresses patient-specific needs rather than serving as a scalable manufacturing pathway. In this study, both formulations used semaglutide base from

FDA-registered manufacturers, meaning the active pharmaceutical ingredient was chemically identical. This eliminates concerns about peptide form (base vs. salt) affecting pharmacologic comparability and supports the validity of direct outcome comparisons. The observed equivalence in weight loss and similar safety profile suggest that when quality-controlled compounding practices are followed, the therapeutic performance of compounded semaglutide can match that of the branded product.

The FDA's concerns about compounded medications center on quality assurance, consistency, and the lack of formal approval processes [9]. Our findings, which specifically evaluated a standardized compounded formulation from a single pharmacy partner, cannot be extrapolated to all compounding practices. However, they demonstrate that when quality control measures are in place, compounded semaglutide can deliver outcomes comparable to branded therapy. This suggests a potential pathway for ensuring access to effective obesity treatment while maintaining appropriate safety oversight.

Strengths and Limitations

This study's strengths include its large sample size, real-world setting, prespecified equivalence framework, and comprehensive statistical approach including propensity score matching and multiple sensitivity analyses (Supplementary Table S2, Supplementary Figure S3). The use of a single compounding pharmacy reduces variability and increases the generalizability of findings to similar standardized compounding practices.

This study's observational design is both a

strength and a limitation. While it reflects real-world clinical practice, it introduces potential for unmeasured confounding that randomization would eliminate. Although we adjusted for measured confounders (age, sex, BMI) and conducted propensity score matching, patients self-selected (or were channeled by insurance or cost factors into) treatment groups. Unmeasured factors (such as motivation, health literacy, or baseline comorbidity profiles) may differ between groups and influence both treatment choice and outcomes. The consistency of findings across adjusted, matched, and sensitivity analyses (Supplementary Table S2, Supplementary Figure S3) reduces but does not eliminate this concern. The analysis is limited to 20 to 28 weeks of follow-up; longer-term outcomes remain to be established. Safety comparisons were underpowered due to the small, branded safety sample and passive adverse event capture (Supplementary Table S4). The study population was predominantly female and accessed care through telemedicine, which may limit generalizability to other care settings. Finally, the use of different formulations (semaglutide base in both products, co-formulated with cyanocobalamin in CSC) means we cannot isolate effects of compounding from potential effects of cyanocobalamin, though the therapeutic equivalence observed suggests any such effects are negligible for weight outcomes.

Conclusions

Summary of Findings

This observational study of nearly 5,000 patients provides compelling evidence that compounded semaglutide plus cyanocobalamin achieves weight loss statistically and clinically equivalent to branded Wegovy over 20 to 28 weeks. The adherence advantage of CSC was consistent across all demographic and clinical subgroups, suggesting robust generalizability. Safety profiles appeared similar between treatments, though formal equivalence testing was limited by sample size constraints in the branded comparator group. These findings support expanded access to obesity pharmacotherapy during a critical period when effective treatments exist but remain financially inaccessible to many who would benefit. The challenge for healthcare systems and regulators is ensuring that expanded access through compounding maintains the quality standards necessary to protect patient safety while addressing the urgent public health need for scalable obesity treatment. Our results demonstrate that when appropriate quality controls are in place, this balance is achievable.

Clinical and Policy Implications

These findings have direct implications for clinical practice and healthcare policy. For the first time, rigorous comparative effectiveness research demonstrates that standardized compounded semaglutide from quality-controlled sources delivers weight-loss outcomes equivalent to the FDA-approved branded product while significantly improving treatment persistence. This combination of therapeutic equivalence and superior adherence positions CSC as more than simply an alternative during shortages. Rather, it represents a viable long-term treatment option that may improve real-world outcomes through sustained medication use.

The clinical significance extends beyond individual patient care. Obesity affects over 40% of U.S. adults, yet access to effective pharmacotherapy remains severely limited by cost barriers. Branded semaglutide costs $900 to $1,500 monthly without insurance coverage, placing it beyond reach for millions of eligible patients. Compounded semaglutide at $200 to $400 monthly substantially expands the population that can access and maintain GLP-1 therapy. Given that obesity is a chronic disease requiring sustained treatment, the observed adherence advantage with CSC could translate to meaningfully better long-term outcomes at the population level.

For clinicians, these results provide evidence-based support for prescribing standardized compounded semaglutide when patients face financial barriers to branded therapy or when supply constraints limit access. The key qualifier is standardization: our findings apply specifically to compounded formulations prepared by pharmacies following rigorous quality assurance protocols, using FDA-registered peptide sources, and maintaining batch-to--batch consistency through systematic testing. These findings should not be extrapolated to all compounding practices without similar quality controls.

For policymakers, this research demonstrates that well-regulated compounding can provide a pathway to ex-

panded access without compromising therapeutic outcomes. The challenge lies in establishing and enforcing quality standards that ensure patient safety while maintaining the flexibility that makes compounding valuable during drug shortages or for patients with individual clinical needs. Our findings suggest that when appropriate oversight mechanisms are in place, compounded semaglutide can serve as an effective tool for addressing the obesity epidemic while medication access remains constrained.

Study Limitations and Future Directions

Several important limitations temper these conclusions. First, the observational design cannot definitively rule out unmeasured confounding despite extensive adjustment, propensity score matching, and multiple sensitivity analyses. While our findings were consistent across all analytical approaches, residual confounding from unmeasured factors (such as patient motivation, health literacy, or specific comorbidity patterns) may influence both treatment selection and outcomes. Randomized controlled trials would provide more definitive evidence of equivalence.

Second, follow-up was limited to 20 to 28 weeks (approximately 5 to 6 months). Weight maintenance beyond this period remains uncharacterized, as do long-term cardiovascular and metabolic outcomes. Given that obesity treatment typically requires years of sustained therapy, extended follow-up is critical to assess durability of weight loss, long-term safety, and whether the early adherence advantage with CSC persists over time.

Third, safety comparisons were substantially underpowered due to the small branded comparator sample (n= 42) and reliance on passive adverse event capture through routine clinical documentation. While we observed no concerning safety signals and several adverse events were numerically less common with CSC, these findings are descriptive rather than confirmatory. Prospective studies with systematic, adjudicated adverse event surveillance and adequate sample sizes in both treatment arms are needed to establish formal safety equivalence.

Fourth, our findings apply specifically to a standardized compounded formulation from a single pharmacy partner using defined quality assurance protocols. Generalizability to other compounding practices depends on comparable quality control measures, including use of pharmaceutical-grade semaglutide base from FDA-registered suppliers, systematic purity and sterility testing, and consistent manufacturing processes. The study population (predominantly female, telemedicine-based care) may also limit generalizability to other demographic groups and care settings.

Future research should address these gaps through: (1) longer-term comparative effectiveness studies extending to 12 to 24 months or beyond; (2) prospective trials with systematic safety surveillance and adequate power for equivalence testing; (3) cardiovascular and metabolic outcome studies to assess whether therapeutic equivalence extends beyond weight loss to diabetes prevention, blood pressure control, and cardiovascular risk reduction; (4) economic analyses quantifying the population health impact of expanded access through lower-cost compounded alternatives; and (5) quality assessment studies characterizing variation across compounding pharmacies to inform regulatory standards.

Funding

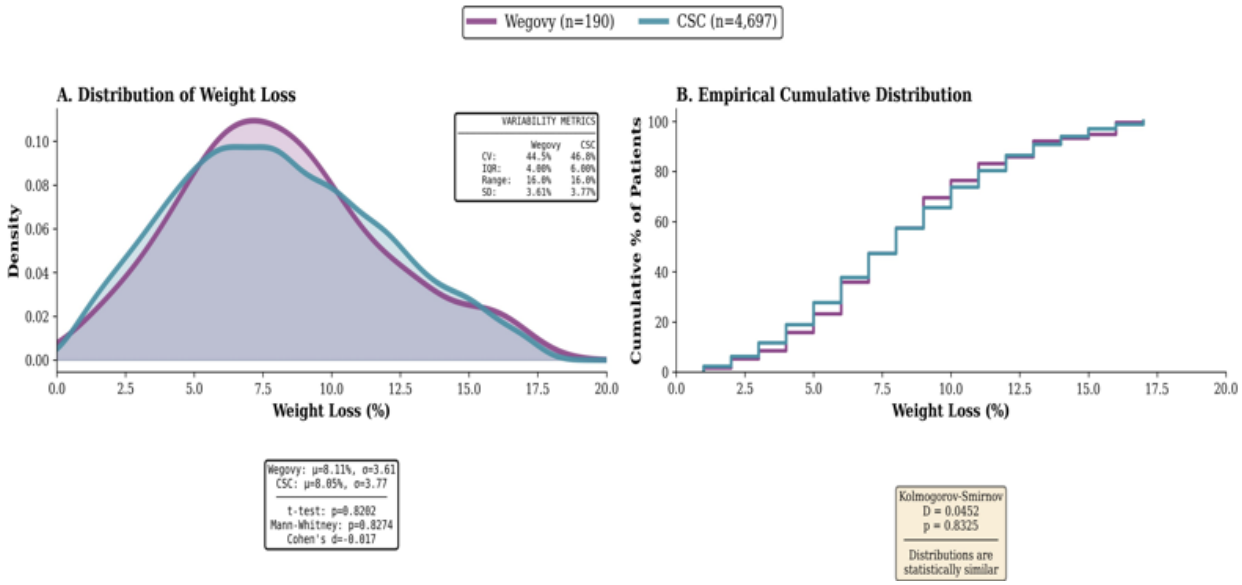
No external funding was received for this work. The study was investigator-initiated and conducted using de-identified operational data provided by Mochi Health. Mochi Health had no role in the study design, data analysis, data interpretation, manuscript preparation, or the decision to submit the manuscript for publication.

Disclosures

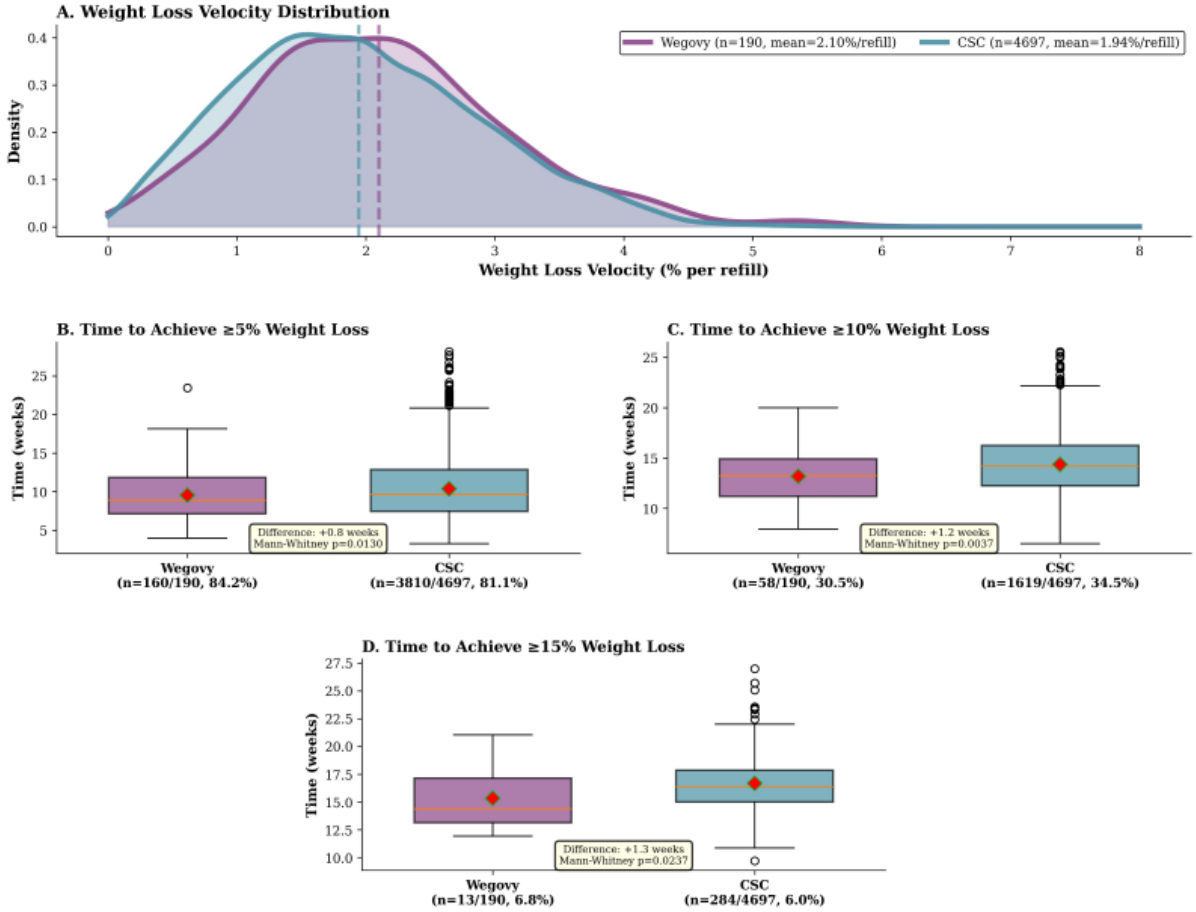
Shanmugesh Raja is an employee/contractor of Mochi Health.

Supplementary Materials

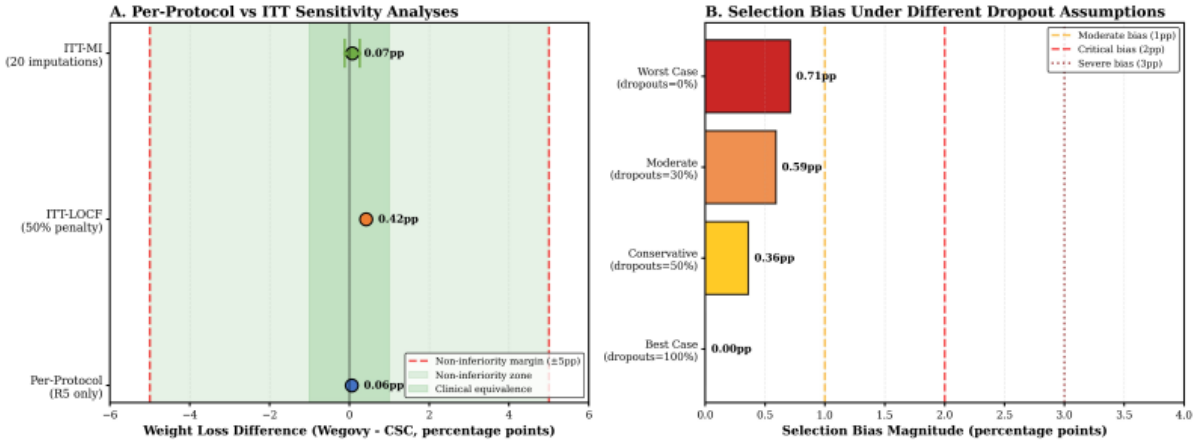
Supplementary Analyses



Supp. Figure. S1: Distribution of weight-loss responses. *Left*: smoothed density curves of percent weight loss at R5 for WGV and CSC. Right: empirical cumulative distribution functions with KolmogorovSmirnov test (D=0.0452, p=0.832).



Supp. Figure. S2: Weight-loss velocity and time-to-response. *Panel A*: refill-to-refill percent weight change by medication. Panels B to D: time to achievement of ≥5%, ≥10%, and ≥15% weight loss with Mann-Whitney comparisons (all $p > 0.05$).



Supp. Figure. S3: Intention-to-treat sensitivity analyses. *Panel A*: Per-protocol vs. ITT estimates with equivalence bounds. Panel B: Selection-bias scenarios under various dropout assumptions.

Table S1: Mediation analysis: persistence as mediator

Mediation Step	Coefficient	P Value
BARON KENNY CAUSAL STEPS		
Total effect (c): Medication → Weight loss	0.063	0.82
Path a: Medication → Persistence	-0.691 (OR = 0.501)	<0.001
Path b: Persistence → Weight loss	0.328	0.008
Direct effect (c'): Medication → Weight loss (adjusted for persistence)	0.099	0.722
MEDIATION SUMMARY		
Indirect effect (a × b)	-0.227	—
Sobel test	Z = -2.074	0.038

Supp. Table. S1: Mediation analysis. Baron-Kenny paths showing medication influences weight loss indirectly through adherence.

Table S2: Propensity score matching and multiple testing correction

Analysis	Before Matching	After Matching
PROPENSITY SCORE MATCHING		
Sample size—CSC	n = 4,697	n = 190
Sample size—Wegovy	n = 190	n = 190
COVARIATE BALANCE (SMD)		
BMI	0.026	0.012
Female sex	−0.087	−0.144
Duration to R5	−0.605	0.03
Age	−0.323	−0.025
TREATMENT EFFECT (MATCHED)		

Wegovy mean weight loss, %	—	8.11
CSC mean weight loss, %	—	8.06
Average treatment effect, pp	—	0.05 (95% CI −0.69, 0.80)
P value	—	0.89
MULTIPLE TESTING CORRECTION		
Total hypothesis tests	13	—
Bonferroni-significant findings	Age, Duration, Persistence	—
Benjamini–Hochberg significant	Age, Duration, Persistence	—
Weight loss p value (adjusted)	p = 0.883	—

PS covariates: BMI, sex, duration to R5, age. 1:1 nearest-neighbor matching without replacement.

SMD <0.1 indicates good balance. ATE from matched sample; bootstrapped 95% CI.

FDR control at α=0.05 using Benjamini–Hochberg procedure.

Abbreviations: PS, propensity score; SMD, standardized mean difference; ATE, average treatment effect; pp, percentage points; CI, confidence interval; FDR, false discovery rate.

Supp. Table. S2: Propensity score matching and multiple testing. Matched-sample analysis confirms primary findings; multiple comparison adjustments reported.

Table S3: Weight Loss Outcomes by Ethnicity

Ethnicity	Wegovy N (%)	Weight Loss, % Mean ± SD	CSC N (%)	Weight Loss, % Mean ± SD	Difference	P Value
Non-Hispanic White	114 (60.0)	8.22 ± 3.50	3414 (72.7)	8.21 ± 3.76	–0.01%	0.986
Hispanic	33 (17.4)	8.00 ± 4.18	364 (7.7)	7.71 ± 3.81	–0.29%	0.674
Black	22 (11.6)	7.41 ± 3.47	290 (6.2)	7.65 ± 3.84	0.0024	0.774
Other	4 (2.1)	8.50 ± 4.36	204 (4.3)	7.44 ± 3.68	—	—
Asian	6 (3.2)	8.33 ± 2.58	139 (3.0)	7.17 ± 3.48	–1.16%	0.422
Multiracial / Multi-ethnic	3 (1.6)	7.00 ± 3.61	130 (2.8)	7.61 ± 3.96	—	—
All patients	190 (100.0)	8.11 ± 3.61	4697 (100.0)	8.05 ± 3.77	–0.06%	0.82

P values from independent t-tests comparing weight loss between medications within each ethnic group.Only groups with n ≥ 5 in both cohorts are included in statistical testing.

Supp. Table. S3: Weight-loss outcomes by ethnicity. No significant between-medication differences observed in groups with adequate sample size.

Table S4: Adverse events and safety profile

Adverse Event	CSC per 100	Wegovy per 100	P Value

ANY REPORTED AE	180.2	106.8	—
Specific Adverse Events			
GI/Nausea	0	0	—
Fatigue	10.8	4.7	0.011
Headache	8.9	3.2	0.009
Heartburn	6.6	3.7	0.153
Acid reflux	4	1.1	0.063
Severity			
Mild	132.4	83.2	—
Moderate	73.9	40	—
Severe	12.3	4.7	—

Rates calculated per 100 refills.

Safety analysis subcohort: CSC n = 956; Wegovy n = 42.

P values from χ^2 tests where available; exploratory analysis—not powered for non-inferiority.

Abbreviations: AE = adverse event; GI = gastrointestinal; CI = confidence interval.

Supp. Table. S4: Adverse events by symptom type. Item-level comparison of specific adverse events between treatment groups.

Table 4: Subgroup analysis by age, BMI, and sex

Subgroup	CSC n	CSC Mean ± SD	Wegovy n	Wegovy Mean ± SD	P Value
AGE <40 years	2441	8.30 ± 3.79	119	8.61 ± 3.58	0.383
AGE 40–49 years	1142	7.84 ± 3.79	43	7.14 ± 3.25	0.23
AGE 50–59 years	698	7.78 ± 3.79	23	6.96 ± 3.94	0.307
AGE ≥60 years	416	7.60 ± 3.44	5	10.00 ± 3.46	0.121
BMI <30 kg/m²	399	8.29 ± 3.74	16	8.81 ± 3.27	0.583
BMI 30–34.9 kg/m²	1826	8.60 ± 3.79	74	8.39 ± 3.59	0.646
BMI 35–39.9 kg/m²	1286	7.86 ± 3.70	48	7.83 ± 3.45	0.968
BMI ≥40 kg/m²	1186	7.33 ± 3.68	52	7.75 ± 3.91	0.416
Sex Female	3985	8.13 ± 3.81	155	8.00 ± 3.61	0.684
Sex Male	712	7.60 ± 3.48	35	8.60 ± 3.62	0.099

Wegovy n < 5: descriptive only.BMI categories as per WHO classification.Treatment×BMI interaction p = 0.412; Treatment×Sex interaction p = 0.183 (not significant).P values from Welch t-tests; unadjusted.Abbreviations: **BMI**, body mass index; **n.s.**, not significant; **WHO**, World Health Organization.

Supp. Table. S5: BMI subgroup outcomes. Comparative effectiveness consistent across baseline BMI strata (Pinteraction=0.891).

Table S5: Comprehensive Statistical Testing Summary

Statistical Test	Purpose	Test Statistic	P Value	Conclusion
Distribution & Equivalence				
Kolmogorov–Smirnov	Compare distributions	D = 0.0452	0.8325	Distributions equivalent
Central Tendency				
Independent t-test	Compare means	t = –0.227	0.8202	Means not different
Mann–Whitney U	Compare medians	U = 442069	0.8274	Medians not different
Cohen's d	Effect size	d = –0.0168	—	Negligible (<0.2)
Variance				
Levene's test	Compare variances	F = 1.902	0.1679	Equal variances
Responder Rates				
Chi-square (≥5%)	Compare responder rates	$\chi^2 = 0.953$	0.3288	No difference
Chi-square (≥10%)	Compare responder rates	$\chi^2 = 1.090$	0.2964	No difference
Chi-square (≥15%)	Compare responder rates	$\chi^2 = 0.087$	0.7678	No difference
Survival Analysis				
Log-rank test	Compare persistence curves	$\chi^2 = 5.724$	0.0167*	CSC superior persistence
Cox PH (adjusted)	Hazard ratio	HR = 0.662 [0.565–0.775]	<0.001*	CSC 33.8% lower risk

Statistically significant ($P < 0.05$). All tests two-sided unless specified.Cox PH model adjusted for age, gender, and baseline BMI.

Supp. Table. S6: Comprehensive statistical testing matrix. Complete summary of all distributional, central tendency, variance, responder, and survival comparisons.

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